THE GAME

True. False.

Or we don't know yet? 36 questions to discover what science really knows.

01 Prepare

Print and cut out the 36 cards. Shuffle the deck and keep the answers face down.

02 Read

The player who draws reads the question aloud without showing the back. Other players choose: true, false, or we don't know yet.

03 Reveal

Flip the card and read the scientific explanation. Score one point for each correct answer: recognizing what science does not know yet counts too!

04 Play Together

After 10 cards, the player with the most points wins. In team mode, work together to score at least 8 out of 10.

12 TRUE / 12 FALSE / 12 OPEN QUESTIONS

Recommended age: 7-12 years. Every card back displays its number and matching SC reference code.

BEFORE PRINTING

A Clean Print.

4 cards per A4 sheet. 9 sheets for the complete double-sided deck.

Files and Orientation

Open OTTO_CARTE_36_FRONTE_RETRO_A4.pdf. Use portrait A4, print at 100% or 'actual size', and uncheck 'fit to page'.

Double-Sided Printing

Select double-sided printing on the long edge. Backs are already column-swapped so card 01 aligns directly with back 01.

Test Print First

Print pages 1 and 2 first. Hold up to the light to verify alignment. Measure the card: it should be 82 × 118 mm.

Paper and Cutting

Use sturdy matte paper compatible with your printer (e.g. 160-250 gsm). Follow the crop marks with a ruler and cutter or scissors.

Single-Sided Option

Use OTTO_CARTE_36_SOLO_FRONTE_A4.pdf: 9 pages of questions. Answers and explanations are in this guide.

SOLUTIONS

Cards 01-08

First choose an answer. Then check against the scientific explanation.

01 TRUE — SC-01

Yes. Researchers observed mixed hunting parties searching for prey together, with different roles shared between octopus and fish.

02 WE DON'T KNOW YET — SC-03

Some vocal calls seem addressed to specific individual elephants. We don't yet know if they work exactly like human names in every way.

03 FALSE — SC-04

That is not what researchers documented. Some orcas roll pieces of kelp across their bodies as an allogrooming or social behavior.

04 WE DON'T KNOW YET — SC-04

No. Skin care, soothing parasites, and social bonding are interesting hypotheses, but the exact function remains under investigation.

05 TRUE — SC-05

Yes. Solitary and group humpbacks blow circular bubble nets that concentrate fish and krill before lunging upward.

06 FALSE — SC-06

No. The study analyzed crows producing a specific small number of vocal calls matching visual cues from 1 to 4.

07 TRUE — SC-02

Yes! In Sydney, wild cockatoos learned to grip and push fountain handles with their feet and body weight to drink running water.

08 WE DON'T KNOW YET — SC-16

The rolling behavior meets scientific criteria for animal play. We do not know what their internal subjective experience feels like.

SOLUTIONS

Cards 09-16

First choose an answer. Then check against the scientific explanation.

09 TRUE

SC-09

Yes. When dehydrated or cut, tomato and tobacco plants emit ultrasonic popping sounds detectable by specialized microphones.

10 FALSE

SC-13

No. Genome size does not determine an organism's intelligence or complexity. Much of a giant genome consists of repetitive sequences.

11 WE DON'T KNOW YET

SC-09

Acoustic emissions have been measured. Whether they serve as intentional communication signals remains unproven.

12 TRUE

SC-12

Yes, in a lab on Earth using Apollo soil samples. However, the plants showed severe stress responses compared to earthly volcanic ash.

13 TRUE

SC-14

Yes! Nepenthes pudica in Borneo develops underground pitcher traps that capture ants, mites, and subterranean invertebrates.

14 FALSE

SC-11

No. Survival depends strictly on the organism, solar UV exposure, temperature extremes, and duration in space.

15 TRUE

SC-15

Yes. A specialized robotic vehicle tracked the fish's swimming position and translated it into motor commands, allowing it to navigate a room.

16 FALSE

SC-15

No! There was no steering wheel. An overhead camera and computer tracked where the fish faced and moved the wheels accordingly.

SOLUTIONS

Cards 17-24

First choose an answer. Then check against the scientific explanation.

17 **TRUE** SC-16

Yes. Bees rolled balls repeatedly without food incentives, showing that the behavior was intrinsically motivating.

18 **WE DON'T KNOW YET** SC-19

No. Experiments measure hesitation and body posture, but science cannot directly read an animal's subjective thoughts.

19 **FALSE** SC-20

No. It is applied physics and chemistry! Water temperature, starch concentration, and shear mixing determine emulsion stability.

20 **FALSE** SC-23

No. DNA decays over millions of years. We do not have intact T. rex DNA; Jurassic Park remains science fiction.

21 **TRUE** SC-22

Yes! Biologically and anatomically, birds are avian dinosaurs belonging to the theropod lineage.

22 **FALSE** SC-22

No. Chickens and T. rex share distant common theropod ancestors, but T. rex is an extinct cousin, not a direct ancestor.

23 **WE DON'T KNOW YET** SC-21

Thermal insulation, mating displays, and sensory functions are leading hypotheses. Feathers existed long before flight evolved.

24 **WE DON'T KNOW YET** SC-26

Gene editing can alter pigeon genomes to express dodo traits, but whether that recreates the lost species remains an open debate.

SOLUTIONS

Cards 25-32

First choose an answer. Then check against the scientific explanation.

25 WE DON'T KNOW YET

SC-32

Transparent wood is currently developed in labs. Large-scale industrial production, durability, and cost will determine future adoption.

26 FALSE

SC-36

No. Electrodes capture natural bioelectrical signals from fungal threads, and human-engineered circuits convert them into motor commands.

27 TRUE

SC-39

Yes. Erodium seed awns bend and twist hygroscopically when air humidity changes, driving the seed into the ground without a motor.

28 WE DON'T KNOW YET

SC-34

It depends on the task. Bio-inspired soft robotics is powerful for grasping delicate objects, but rigid arms are faster for heavy factory work.

29 TRUE

SC-43

Yes. Neuroprosthetic interfaces record muscular and neural signals to provide intuitive, natural prosthetic joint movement.

30 FALSE

SC-44

No. Retinal implants are designed for specific degenerations and provide artificial light patterns, not restored normal eyesight.

31 WE DON'T KNOW YET

SC-46

Clinical trials decoding brain activity into synthesized speech are highly promising, but broader clinical availability requires more years of testing.

32 TRUE

SC-55

Yes. The Event Horizon Telescope combined synchronized radio data from observatories around the globe using supercomputers.

SOLUTIONS

Cards 33-36

First choose an answer. Then check against the scientific explanation.

33 FALSE

SC-55

No. It is an interferometric reconstruction built from petabytes of radio wave data collected across Earth, not a conventional photo snapshot.

34 WE DON'T KNOW YET

SC-52

No. Interstellar objects come from other star systems unpredictably. Future discoveries depend on new survey sky telescopes like Vera Rubin.

35 FALSE

SC-52

No. 3I/ATLAS has an open, hyperbolic trajectory with high speed: it passes through the solar system once and flies back into interstellar space.

36 WE DON'T KNOW YET

SC-54

Long-duration life support, radiation shielding, energy, and transportation must first be tested and proven. No certain date exists.

SOURCES & EVIDENCE

Follow the Evidence.

SC codes link directly to the book sources and scientific publications.

The 36 questions explore discoveries highlighted in the book. Each card connects to genuine peer-reviewed studies or institutional scientific records.

The illustrations are reconstructed visualizations created for the book. The M87* image on cards 32 and 33 is the official photographic reconstruction: EHT Collaboration / ESO (2019), CC BY 4.0.

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Card size: 82 × 118 mm. Margins: 18 mm left/right, 24.5 mm top/bottom. Grid spacing: 10 mm horizontally, 12 mm vertically. Trim marks are placed safely outside the card area.